# Statistics 280 — Practice Midterm 2A

Unofficial practice exam in the style of the W25 and W26 Midterm 2.
Covers Lectures 9–15 (Chapters 13–17): probability, Bayes' rule, counting, binomial and hypergeometric distributions, random variables, expected value, standard error, covariance, box models, and the normal approximation, plus R for probability and simulation.

Name _____ Start time / End time _____

- Answer all questions in the space provided and indicate where you have written additional work.
- Solutions without justification will be marked as wrong for all questions except for questions asking for R code. Show all work. A calculator command does not count as justification.
- Question 7 is an extra credit question (15 points).
- Express answers as fractions or as decimals rounded to 2 decimal places.
- The last page is a table of values from the normal distribution. Closed book, no notes, calculator allowed.
- Suggested time: 80 minutes. Total: 200 points + 15 extra credit.

| **Question** | **1** | **2** | **3** | **4** | **5** | **6** | **7** | **Total** |
|---|---|---|---|---|---|---|---|---|
| Points | 30 | 30 | 30 | 40 | 30 | 40 | 15 | 200 |
| Score | | | | | | | | |

**1. (30 points)** An escape room has a ring of 12 keys, and exactly 2 of them open the door. You try keys one at a time, chosen at random, and set aside each key that fails, so you never try a key twice.

**a)** (5 points) What is the chance that the first key you try opens the door?

**b)** (6 points) What is the chance that the door opens for the first time on exactly your 3rd try?

**c)** (6 points) What is the chance that you need more than 4 tries to open the door?

**d)** (7 points) What is the chance that both working keys are among the first 5 keys you try?

**e)** (6 points) Before starting, you lay all 12 keys in a random order on a table and plan to try them left to right. What is the chance that the 7th key in the row opens the door? Explain why the answer doesn't depend on the position.

**2. (30 points)** A dorm has 200 residents. 160 are taking a statistics class (S), 150 are taking a computer science class (C), and 140 play an intramural sport (I).

**a)** (6 points) What are the maximum and minimum possible numbers of residents taking both statistics and computer science? Describe the situation that gives each.

**b)** (6 points) If taking statistics and taking computer science were independent, how many residents would take both? How many would take statistics or computer science?

**c)** (8 points) What are the maximum and minimum possible numbers of residents who are in all three groups (S, C and I)? Justify the minimum.

**d)** (4 points) Could S and C be mutually exclusive here? Can two events that each have positive probability be both independent and mutually exclusive? Explain.

**e)** (6 points) Suppose 112 residents take both statistics and computer science. Find P(C | S) and P(S | C) for a randomly chosen resident. Are S and C independent? Give a frequency interpretation of P(C | S).

**3. (30 points)** An email provider estimates that 20% of incoming email is spam. The word "winner" appears in 40% of spam emails and in 2% of legitimate emails.

**a)** (4 points) What are the prior odds that an incoming email is spam?

**b)** (4 points) What are the data odds for spam given that an email contains "winner"?

**c)** (6 points) What are the posterior odds that an email containing "winner" is spam? What is the probability that it is spam?

**d)** (6 points) Spam is also sent between 2 and 4 a.m. 30% of the time, while legitimate email is sent then 5% of the time. Assume that, within spam and within legitimate email, the time of sending is independent of whether the email contains "winner". What is the probability that an email containing "winner" and sent at 3 a.m. is spam?

**e)** (5 points) What percentage of incoming email would need to be spam for an email containing "winner" to have exactly a 50% chance of being spam?

**f)** (5 points) What percentage of all incoming emails contain the word "winner"? Give a frequency interpretation of your answer to part c).

**4. (40 points)** A jury pool of 40 people contains 12 people under the age of 30. A jury of 12 is chosen at random from the pool.

**a)** (8 points) What is the chance that exactly 2 of the 12 jurors are under 30? Write the answer with choose notation and evaluate it.

**b)** (4 points) Write the R command to calculate the chance in part a). Include all inputs.

`R command:` _____

**c)** (4 points) Write the R command to calculate the chance that at most 1 juror is under 30.

`R command:` _____

**d)** (5 points) Write the R command to calculate the chance that at least 4 jurors are under 30.

`R command:` _____

**e)** (7 points) In a large county, 30% of eligible jurors are under 30. Treat the 12 jurors as independent draws from the county. What is the chance that exactly 2 jurors are under 30?

**f)** (4 points) In the setting of part e), write the R command for the chance that between 2 and 5 jurors, inclusive, are under 30.

`R command:` _____

**g)** (4 points) In the setting of part e), give the expected value and standard error of the number of jurors under 30.

**h)** (4 points) Why is the binomial distribution not appropriate for part a), even though the fraction of the pool under 30 (12/40) is also 30%?

**5. (30 points)** You play the following game. Roll a fair die. If it shows 1 or 2, you pay $3. If it shows 3, 4, or 5, you win $1. If it shows 6, you roll again and win the number of dollars shown on the second roll. Let X be your net winnings from one game.

**a)** (8 points) Give a table of the possible values of X and their probabilities.

**b)** (8 points) Find E(X) and SE(X).

**c)** (7 points) Write an R function `play()` that simulates one game and returns X.

**d)** (7 points) Write R code that uses `play()` to estimate E(X), SE(X), and P(X > 0) from 100,000 simulated games.

**6. (40 points)** A basketball player makes 80% of her free throws, independently from shot to shot. She takes 400 free throws over a season.

**a)** (6 points) Describe a box model for the number of free throws she makes: the tickets, the average and SD of the box, and the number of draws. Find the expected value and standard error of the number she makes.

**b)** (5 points) Find the expected value and standard error of the percentage of free throws she makes.

**c)** (5 points) Write the R command for the exact chance that she makes at least 330 free throws.

`R command: _____`

**d)** (10 points) Use the normal approximation, with the continuity correction, to estimate the chance that she makes at least 330 free throws.

**e)** (6 points) Without calculating, which is more likely: that she makes at least 85% of her free throws in a 100-shot stretch, or at least 85% in the 400-shot season? Explain using the law of averages and standard errors.

**f)** (8 points) She makes a deal with a friend: she wins $1 for each made free throw and pays $3 for each miss. Use the short-cut formula for the SD of a box with two values to find the expected value and standard error of her net winnings over the 400 shots. Then estimate the chance that she loses money overall (ignore the continuity correction).

**7. (15 points extra credit)**

Suppose that events A and B are independent. Show that A and not-B are also independent.

# A Normal Table

Height = height of the standard normal curve at z (percent per standard unit). Area = percentage of the area under the normal curve between −z and z.

| z | Height | Area | z | Height | Area | z | Height | Area |
|---|---|---|---|---|---|---|---|---|
| **0.00** | 39.89 | 0.00 | **1.50** | 12.95 | 86.64 | **3.00** | 0.443 | 99.730 |
| **0.05** | 39.84 | 3.99 | **1.55** | 12.00 | 87.89 | **3.05** | 0.381 | 99.771 |
| **0.10** | 39.70 | 7.97 | **1.60** | 11.09 | 89.04 | **3.10** | 0.327 | 99.806 |
| **0.15** | 39.45 | 11.92 | **1.65** | 10.23 | 90.11 | **3.15** | 0.279 | 99.837 |
| **0.20** | 39.10 | 15.85 | **1.70** | 9.40 | 91.09 | **3.20** | 0.238 | 99.863 |
| **0.25** | 38.67 | 19.74 | **1.75** | 8.63 | 91.99 | **3.25** | 0.203 | 99.885 |
| **0.30** | 38.14 | 23.58 | **1.80** | 7.90 | 92.81 | **3.30** | 0.172 | 99.903 |
| **0.35** | 37.52 | 27.37 | **1.85** | 7.21 | 93.57 | **3.35** | 0.146 | 99.919 |
| **0.40** | 36.83 | 31.08 | **1.90** | 6.56 | 94.26 | **3.40** | 0.123 | 99.933 |
| **0.45** | 36.05 | 34.73 | **1.95** | 5.96 | 94.88 | **3.45** | 0.104 | 99.944 |
| **0.50** | 35.21 | 38.29 | **2.00** | 5.40 | 95.45 | **3.50** | 0.087 | 99.953 |
| **0.55** | 34.29 | 41.77 | **2.05** | 4.88 | 95.96 | **3.55** | 0.073 | 99.961 |
| **0.60** | 33.32 | 45.15 | **2.10** | 4.40 | 96.43 | **3.60** | 0.061 | 99.968 |
| **0.65** | 32.30 | 48.43 | **2.15** | 3.96 | 96.84 | **3.65** | 0.051 | 99.974 |
| **0.70** | 31.23 | 51.61 | **2.20** | 3.55 | 97.22 | **3.70** | 0.042 | 99.978 |
| **0.75** | 30.11 | 54.67 | **2.25** | 3.17 | 97.56 | **3.75** | 0.035 | 99.982 |
| **0.80** | 28.97 | 57.63 | **2.30** | 2.83 | 97.86 | **3.80** | 0.029 | 99.986 |
| **0.85** | 27.80 | 60.47 | **2.35** | 2.52 | 98.12 | **3.85** | 0.024 | 99.988 |
| **0.90** | 26.61 | 63.19 | **2.40** | 2.24 | 98.36 | **3.90** | 0.020 | 99.990 |
| **0.95** | 25.41 | 65.79 | **2.45** | 1.98 | 98.57 | **3.95** | 0.016 | 99.992 |
| **1.00** | 24.20 | 68.27 | **2.50** | 1.75 | 98.76 | **4.00** | 0.013 | 99.9937 |
| **1.05** | 22.99 | 70.63 | **2.55** | 1.54 | 98.92 | **4.05** | 0.011 | 99.9949 |
| **1.10** | 21.79 | 72.87 | **2.60** | 1.36 | 99.07 | **4.10** | 0.009 | 99.9959 |
| **1.15** | 20.59 | 74.99 | **2.65** | 1.19 | 99.20 | **4.15** | 0.007 | 99.9967 |
| **1.20** | 19.42 | 76.99 | **2.70** | 1.04 | 99.31 | **4.20** | 0.006 | 99.9973 |
| **1.25** | 18.26 | 78.87 | **2.75** | 0.91 | 99.40 | **4.25** | 0.005 | 99.9979 |
| **1.30** | 17.14 | 80.64 | **2.80** | 0.79 | 99.49 | **4.30** | 0.004 | 99.9983 |
| **1.35** | 16.04 | 82.30 | **2.85** | 0.69 | 99.56 | **4.35** | 0.003 | 99.9986 |
| **1.40** | 14.97 | 83.85 | **2.90** | 0.60 | 99.63 | **4.40** | 0.002 | 99.9989 |
| **1.45** | 13.94 | 85.29 | **2.95** | 0.51 | 99.68 | **4.45** | 0.002 | 99.9991 |